

- Brown, G., Bun, M., Feldman, V., Smith, A., and Talwar, K. When is memorization of irrelevant training data necessary for high-accuracy learning? In *Proceedings of the 53rd annual ACM SIGACT symposium on theory of computing*, pp. 123–132, 2021.
- Carlini, N., Liu, C., Úlfar Erlingsson, Kos, J., and Song, D. The secret sharer: Evaluating and testing unintended memorization in neural networks, 2019. URL <https://arxiv.org/abs/1802.08232>.
- Carlini, N., Hayes, J., Nasr, M., Jagielski, M., Sehwag, V., Tramèr, F., Balle, B., Ippolito, D., and Wallace, E. Extracting training data from diffusion models, 2023a. URL <https://arxiv.org/abs/2301.13188>.
- Carlini, N., Ippolito, D., Jagielski, M., Lee, K., Tramèr, F., and Zhang, C. Quantifying memorization across neural language models, 2023b. URL <https://arxiv.org/abs/2202.07646>.
- Cohen, E., Kaplan, H., Mansour, Y., Moran, S., Nissim, K., Stemmer, U., and Tsfadia, E. Data reconstruction: When you see it and when you don’t, 2024. URL <https://arxiv.org/abs/2405.15753>.
- Collins, J., Sohl-Dickstein, J., and Sussillo, D. Capacity and trainability in recurrent neural networks, 2017. URL <https://arxiv.org/abs/1611.09913>.
- Cover, T. M. Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers*, EC-14(3):326–334, 1965.
- Curth, A., Jeffares, A., and van der Schaar, M. A u-turn on double descent: Rethinking parameter counting in statistical learning, 2023. URL <https://arxiv.org/abs/2310.18988>.
- Das, D., Zhang, J., and Tramèr, F. Blind baselines beat membership inference attacks for foundation models, 2024. URL <https://arxiv.org/abs/2406.16201>.
- Delétang, G., Ruoss, A., Duquenne, P.-A., Catt, E., Genewein, T., Mattern, C., Grau-Moya, J., Wenliang, L. K., Aitchison, M., Orseau, L., Hutter, M., and Veness, J. Language modeling is compression, 2024. URL <https://arxiv.org/abs/2309.10668>.
- Duan, M., Suri, A., Mireshghallah, N., Min, S., Shi, W., Zettlemoyer, L., Tsvetkov, Y., Choi, Y., Evans, D., and Hajishirzi, H. Do membership inference attacks work on large language models? In *Conference on Language Modeling (COLM)*, 2024.
- Dubey, A. and et al, A. J. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Dwork, C. Differential privacy. In *International Colloquium on Automata, Languages, and Programming*, pp. 1–12. Springer, 2006.
- Feldman, V. Does learning require memorization? a short tale about a long tail. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing*, pp. 954–959, 2020.
- Gardner, E. The space of interactions in neural network models. *Journal of Physics A: Mathematical and General*, 21(1):257–270, 1988.
- Geiping, J., Stein, A., Shu, M., Saifullah, K., Wen, Y., and Goldstein, T. Coercing llms to do and reveal (almost) anything, 2024. URL <https://arxiv.org/abs/2402.14020>.
- Grunwald, P. and Vitányi, P. Shannon information and kolmogorov complexity. *arXiv preprint cs/0410002*, 2004.
- Grunwald, P. and Vitányi, P. Shannon information and kolmogorov complexity, 2004. URL <https://arxiv.org/abs/cs/0410002>.
- Haghifam, M., Negrea, J., Khisti, A., Roy, D. M., and Dziugaite, G. K. Sharpened generalization bounds based on conditional mutual information and an application to noisy, iterative algorithms. *Advances in Neural Information Processing Systems*, 33:9925–9935, 2020.
- Heinzerling, B. and Inui, K. Language models as knowledge bases: On entity representations, storage capacity, and paraphrased queries, 2021. URL <https://arxiv.org/abs/2008.09036>.
- Jayaraman, B. and Evans, D. Are attribute inference attacks just imputation? In *Proceedings of the 2022 ACM SIGSAC Conference on Computer and Communications Security*, pp. 1569–1582, 2022.
- Kajitsuka, T. and Sato, I. Optimal memorization capacity of transformers, 2024. URL <https://arxiv.org/abs/2409.17677>.
- Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., and Amodei, D. Scaling laws for neural language models, 2020. URL <https://arxiv.org/abs/2001.08361>.

- Kolmogorov, A. N. On tables of random numbers. *Sankhyā: The Indian Journal of Statistics, Series A*, 25(4):369–376, 1963. URL <http://www.jstor.org/stable/25049284>. Accessed: 21/12/2010 15:32.
- Kolmogorov, A. N. Three approaches to the quantitative definition of information. *Problems of Information Transmission*, 1(1):1–7, 1965.
- Lee, K., Ippolito, D., Nystrom, A., Zhang, C., Eck, D., Callison-Burch, C., and Carlini, N. Deduplicating training data makes language models better, 2022. URL <https://arxiv.org/abs/2107.06499>.
- Lu, X., Li, X., Cheng, Q., Ding, K., Huang, X., and Qiu, X. Scaling laws for fact memorization of large language models, 2024. URL <https://arxiv.org/abs/2406.15720>.
- Mahdavi, S., Liao, R., and Thrampoulidis, C. Memorization capacity of multi-head attention in transformers, 2024. URL <https://arxiv.org/abs/2306.02010>.
- Maini, P., Jia, H., Papernot, N., and Dziedzic, A. Llm dataset inference: Did you train on my dataset?, 2024. URL <https://arxiv.org/abs/2406.06443>.
- Mireshghallah, F., Uniyal, A., Wang, T., Evans, D., and Berg-Kirkpatrick, T. Memorization in nlp fine-tuning methods, 2022. URL <https://arxiv.org/abs/2205.12506>.
- Nakkiran, P., Kaplun, G., Bansal, Y., Yang, T., Barak, B., and Sutskever, I. Deep double descent: Where bigger models and more data hurt, 2019. URL <https://arxiv.org/abs/1912.02292>.
- Nasr, M., Carlini, N., Hayase, J., Jagielski, M., Cooper, A. F., Ippolito, D., Choquette-Choo, C. A., Wallace, E., Tramèr, F., and Lee, K. Scalable extraction of training data from (production) language models, 2023. URL <https://arxiv.org/abs/2311.17035>.
- Penedo, G., Kydlíček, H., allal, L. B., Lozhkov, A., Mitchell, M., Raffel, C., Werra, L. V., and Wolf, T. The fineweb datasets: Decanting the web for the finest text data at scale, 2024. URL <https://arxiv.org/abs/2406.17557>.
- Prashanth, U. S., Deng, A., O’Brien, K., V, J. S., Khan, M. A., Borkar, J., Choquette-Choo, C. A., Fuehne, J. R., Biderman, S., Ke, T., Lee, K., and Saphra, N. Recite, reconstruct, recollect: Memorization in lms as a multifaceted phenomenon, 2024. URL <https://arxiv.org/abs/2406.17746>.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., and Sutskever, I. Language models are unsupervised multitask learners. 2019.
- Roberts, A., Raffel, C., and Shazeer, N. How much knowledge can you pack into the parameters of a language model?, 2020. URL <https://arxiv.org/abs/2002.08910>.
- Sablayrolles, A., Douze, M., Ollivier, Y., Schmid, C., and Jégou, H. White-box vs black-box: Bayes optimal strategies for membership inference, 2019. URL <https://arxiv.org/abs/1908.11229>.
- Schwarzschild, A., Feng, Z., Maini, P., Lipton, Z. C., and Kolter, J. Z. Rethinking llm memorization through the lens of adversarial compression, 2024. URL <https://arxiv.org/abs/2404.15146>.
- Shannon, C. E. *A Mathematical Theory of Communication*. University of Illinois Press, 1948. Reprint in 1998.
- Shannon, C. E. Prediction and entropy of printed english, Sept 1950.
- Shokri, R., Stronati, M., Song, C., and Shmatikov, V. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, pp. 3–18. IEEE, 2017.
- Shwartz-Ziv, R., Goldblum, M., Bansal, A., Bruss, C. B., LeCun, Y., and Wilson, A. G. Just how flexible are neural networks in practice?, 2024. URL <https://arxiv.org/abs/2406.11463>.
- Steinke, T. and Zakynthinou, L. Reasoning about generalization via conditional mutual information. In *Conference on Learning Theory*, pp. 3437–3452. PMLR, 2020.
- Tänzer, M., Ruder, S., and Rei, M. Memorisation versus generalisation in pre-trained language models, 2022. URL <https://arxiv.org/abs/2105.00828>.
- Xia, M., Artetxe, M., Zhou, C., Lin, X. V., Pasunuru, R., Chen, D., Zettlemoyer, L., and Stoyanov, V. Training trajectories of language models across scales, 2023. URL <https://arxiv.org/abs/2212.09803>.
- Xu, Y., Zhao, S., Song, J., Stewart, R., and Ermon, S. A theory of usable information under computational constraints, 2020. URL <https://arxiv.org/abs/2002.10689>.

S	Params.	Memorized	Expected	Error
4	6.59×10^5	1.73×10^5	1.80×10^5	4.19
8	6.60×10^5	3.54×10^5	3.60×10^5	1.80
16	6.61×10^5	7.15×10^5	7.21×10^5	0.84
32	6.63×10^5	1.44×10^6	1.44×10^6	0.41
64	6.67×10^5	2.29×10^6	2.36×10^6	2.97
128	6.75×10^5	2.36×10^6	2.39×10^6	1.24
256	6.92×10^5	2.44×10^6	2.45×10^6	0.44

Table 3 Model capacity estimates across sequence length S , along with error (%).

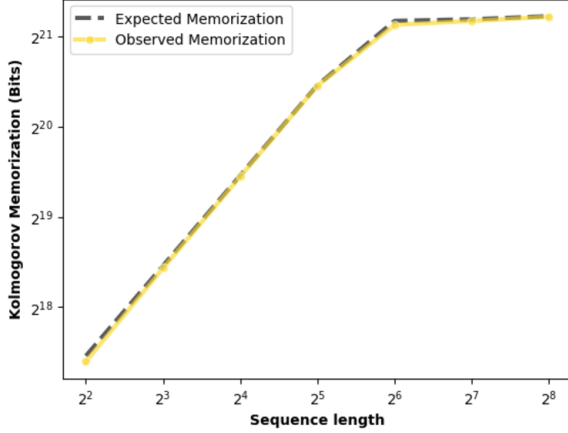


Figure 11 Model memorization across sequence lengths for a fixed-length dataset. Our predictions of total memorization are accurate, with an average error rate of 1.7%.

V	Params.	Memorized	Expected	Error
128	4.21×10^5	1.49×10^6	1.49×10^6	0.36
512	4.71×10^5	1.71×10^6	1.67×10^6	2.78
1024	5.36×10^5	1.95×10^6	1.90×10^6	2.70
2048	6.67×10^5	2.39×10^6	2.36×10^6	1.11
4096	9.29×10^5	3.13×10^6	3.15×10^6	0.47

Table 4 Model capacity estimates across vocab size V , along with error (%).

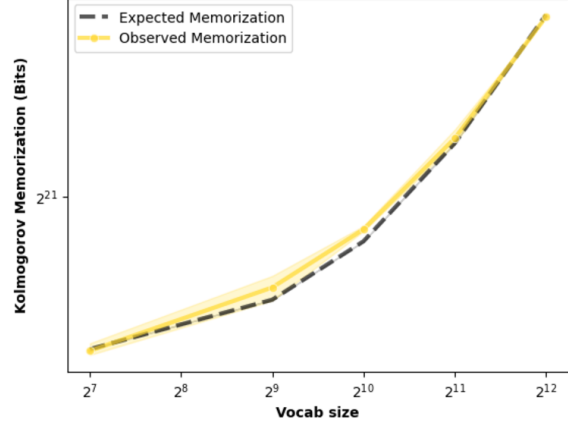


Figure 12 Model memorization across vocabulary size for a fixed-length dataset. Our predictions of total memorization are accurate, with an average error rate of 1.8%. Note that, we do not observe a capacity plateau, since increasing V also increases parameters.

Yeom, S., Giacomelli, I., Fredrikson, M., and Jha, S. Privacy risk in machine learning: Analyzing the connection to overfitting, 2018. URL <https://arxiv.org/abs/1709.01604>.

Yun, C., Sra, S., and Jadbabaie, A. Small relu networks are powerful memorizers: a tight analysis of memorization capacity, 2019. URL <https://arxiv.org/abs/1810.07770>.

Zhang, C., Bengio, S., Hardt, M., Recht, B., and Vinyals, O. Understanding deep learning requires rethinking generalization, 2017. URL <https://arxiv.org/abs/1611.03530>.

Zhang, C., Ippolito, D., Lee, K., Jagielski, M., Tramèr, F., and Carlini, N. Counterfactual memorization in neural language models, 2023. URL <https://arxiv.org/abs/2112.12938>.

A Appendix

A.1 How reliable are our linear estimates of capacity?

Instead of scaling the number of examples in a dataset, we scale model sequence length to adjust the size of a dataset. We use the following measurement for expected memorization of a model:

$$\text{mem}(X, L(X)) \approx \min(\text{capacity}(L), H(X))$$

we substitute our previous estimate of $\alpha = 3.642$ and ensure to adjust the parameter count for increases due to resizing the model’s embedding matrices. We fix the number of training samples to 4096 and train a model

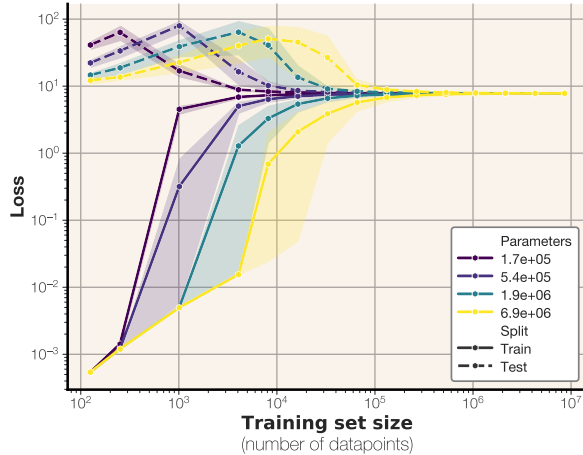


Figure 13 Train and test losses for different-sized language models trained on synthetic data.

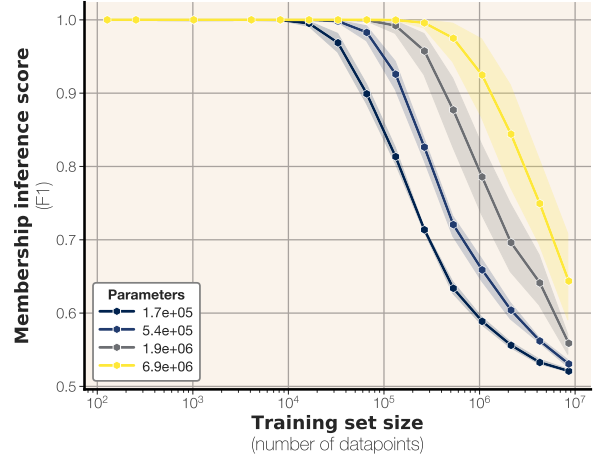


Figure 14 Membership inference attack performance decreases with dataset scale. In the case of uniform synthetic data, membership inference performance never falls below 0.54.

with 2 layers and a hidden size of 128. Results are illustrated in Figure 11 and Table 3. Our predictions of total memorization are accurate, with an average error rate of 1.7% while scaling S and 1.8% when scaling V .

A.2 Additional memorization results

Our findings indicate that memorization of text data neatly plateaus near the model capacity just as in the synthetic data case. When the dataset size increases by a factor of N , the model divides its memorization between datapoints by an equal amount; the sum of memorization is measured to be constant, presumably at the upper bound of the model’s capacity.

When the dataset is small enough for each model to fit – that is, below the capacity of the smallest model – we observe very similar performance between the models. For larger data sizes we notice an interesting trend: unintended memorization increases with dataset size for to a point, presumably as a model fills its capacity with the available information, and then decreases, as the model replaces sample-level information with more useful, generalizable knowledge. A given model generalizes the most (and memorizes the least information about any individual sample) when the dataset is maximally large.

A.3 Comparison of distributions memorized

Distribution-level analysis. Text sequences have very different properties than uniform synthetic bitstrings. We explore how two models of equal capacity spread their memorization across datapoints. We plot a histogram (Figure 15) of train and test compression rates of training data from both synthetic random bitstrings and text. Random training data follows a very normal distribution with a small amount of overlap between train and test compression rates. Text loss is lower on average but more spread out, with low loss on some training points and a long tail of higher losses. There is much more overlap between the train and test loss distributions, which explains why membership inference is more difficult for text data.

Which datapoints are most memorized? Our distribution-level analysis indicates that unlike in the random-bitstring case, models trained on a large amount of text are able to memorize a small number of datapoints. Prior work has indicated that a large amount of this memorization can be due to duplicated training points (Lee et al., 2022) but our dataset is fully deduplicated so this cannot be an explanation in our case.

To quantitatively evaluate the number of rare words per document, we measure the TF-IDF of each training document, plotted vs. unintended memorization in Figure 16. We use the following equation for TF-IDF: